Cameron Zuschmidt

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Science Expected May 2029
B.S.E. in Electrical Engineering, Minor in Computer Science | GPA: 3.94/4.00 *Philadelphia, PA*
Relevant Coursework: Circuits and Systems, Computer System Architecture, Solid State Devices, Electricity and Magnetism, Programming Languages and Techniques

EXPERIENCE

Undergraduate Researcher – Hardware Verification 05/2026 – Present
Penn Instrumentation Group, High-Energy Physics *Philadelphia, PA*

- Perform production quality-control testing on the Power and Timing Card (PTC), a Penn-designed readout board for the DUNE liquid-argon neutrino detector; validating all 250 production boards
- Debug I2C communication failures through MAX7357 mux trees to LTC2945 power/temperature monitors; support Zynq UltraScale+ SoC/FPGA bring-up under PetaLinux
- Develop Python test automation (WebSocket client/server architecture) covering PTC power, timing, sensor, and FPGA checks; contribute to the lab's automated board QC pipeline
- Configure a Beckhoff CX8110 EtherCAT master in TwinCAT and decode EtherCAT process data during board verification
- Authored technical documentation of the PTC architecture and DUNE readout chain; presented a QC research poster

Executive Director and Operations Lead 06/2026 – Present
Penn Aerospace Club *Philadelphia, PA*

- Lead a 150+ member club across jet propulsion, rocketry, high-altitude balloon, and aircraft design teams; manage a $25,000+ budget, including fund allocation and procurement
- Organize speaker events, alumni panels, and industry mentorship connecting members with corporate recruiters

PROJECTS

Avionics Project Lead, High-Power Rocketry Team (IREC) 09/2025 – Present

- Lead avionics for the team's competition rocket; designing a custom STM32-based flight computer PCB in Altium/KiCad with IMU fusion, magnetometers, barometer, radio, GPS, and flash logging, running RTOS firmware
- Implementing a Multiplicative Extended Kalman Filter (MEKF) for quaternion attitude estimation, fusing gyroscope and accelerometer data with inverse-variance weighting
- Integrated a dual-deploy avionics bay (commercial altimeters, GPS, ejection charges); driving control and airbrake actuation

Avionics & Flight Computer Design, High-Altitude Balloon Team 09/2026 – Present

- Designing a custom flight computer PCB that collects and logs all payload sensor data (9-DOF IMU, GPS, pressure, temperature, humidity, UV) for post-flight trajectory reconstruction and weather trajectory analysis

FPGA-Based Digital Control of a Synchronous Buck Converter (In Progress) 08/2026 – Present

- Defined specs (12 V in, 5 V out, 3 A, 150 kHz) and simulating the synchronous buck power stage in LTspice
- Next: Verilog control loop on a Spartan-7 FPGA with PWM generation, dead-time insertion, synchronized ADC sampling, and digital compensator for closed-loop voltage regulation

Hybrid/Liquid Propulsion Initiative 10/2026

- Design teams for the first fully functional hybrid/liquid rocket engines in Penn history, assisting in preliminary systems architecture and feed-system instrumentation for a joint Penn Rocketry and Penn Jet Propulsion initiative

Quantum Entanglement Verification Experiment, MIT iQuHACK 01/2026

- Built a Qiskit circuit to prepare a 20-qubit GHZ (Greenberger–Horne–Zeilinger) state using a Hadamard gate and a CNOT chain, executed on IQM superconducting cloud hardware during a 24-hour hackathon

LEADERSHIP & ACTIVITIES

Avionics Project Lead, Penn High-Power Rocketry (IREC) | **Executive Director and Operations Lead**, Penn Aerospace Club
Teaching Assistant, CIS 1100 (Python) | **Board**, Penn Quantum Computing | **Treasurer**, ESE Student Advisory Board

SKILLS

Languages: Python, C, Verilog, Java
Hardware & EDA: Hardware testing and debugging, Altium Designer, KiCad, LTspice, schematic capture, PCB design
Embedded: STM32, Arduino, Zynq UltraScale+, PetaLinux, RTOS firmware, TwinCAT
Protocols & Tools: I2C, SPI, UART, EtherCAT, Ethernet, LoRa; Git/GitHub, Linux, Bash